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REVIEW

Technical analysis, fundamental analysis, and momentum strategies: Review and integration

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Abstract
This study provides a comprehensive overview of momentum strategies, with an emphasis on integrating both technical and fundamental analysis. It reviews major forms of momentum strategies, including time-series momentum and cross-sectional momentum, and synthesizes empirical evidence on their performance across a wide range of financial markets. The sources of the momentum effect, encompassing both efficient and behavioral explanations, are also discussed. Overall, the literature indicates that momentum strategies remain a widely adopted and economically significant for many market participants. This study concludes by highlighting potential directions for future research on momentum and its underlying mechanisms.

Keywords Fundamental analysis · Market efficiency · Momentum effect · Momentum Strategies · Technical analysis · Trading strategies

JEL Classification G11 · G14

1 Introduction

Investors and financial analysts have long been intrigued by the pursuit of profitable trading strategies. Among the wide array of approaches studied over the decades, technical analysis, fundamental analysis, and momentum-based strategies have emerged as particularly

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prominent. Technical analysis operates on the premise that historical price and volume data contain predictive information about future price movements. In contrast, fundamental analysis emphasizes the intrinsic valuation of securities through the examination of corporate financial statements and macroeconomic indicators, aiming to uncover mispriced assets. Momentum strategies, meanwhile, are predicated on the empirical observation that assets with superior past performance tend to continue outperforming in the future.

This paper centers on momentum strategies, a topic that has garnered substantial attention within academic finance. First documented by Jegadeesh and Titman (1993), the momentum effect has since been robustly validated across diverse asset classes, temporal horizons, and geographic markets, and consistently producing abnormal returns that cannot be fully reconciled with traditional risk-based asset pricing models such as those proposed by Fama and French (1996). The literature has delineated various forms of momentum—including cross-sectional momentum, time-series momentum, earnings momentum, and 52-week high momentum—and continues to investigate their persistence, underlying sources, and interactions with firm-specific attributes and macroeconomic conditions. Despite the wealth of empirical support, the foundational economic mechanisms remain contested. Competing explanations span from delayed information diffusion and investor underreaction to a suite of behavioral biases such as herding behavior, limited investor attention, and overconfidence.

This review paper aims to synthesize the growing body of research on momentum strategies, offering a comprehensive and critical assessment of their empirical performance, theoretical foundations, and practical applications. Specifically, we examine the taxonomy of momentum strategies, the evolution of theoretical explanations—spanning both rational and behavioral frameworks—and the integration of momentum with complementary strategies such as fundamental screening, contrarian positioning, and dynamic asset allocation. While previous studies (e.g., Lee et al. 2024) have touched upon aspects of momentum, few have provided a unified and detailed exposition of its implications across different domains.

This paper contributes to the existing body of knowledge in several key respects. First, by outlining a range of investment approaches—including technical, fundamental, contrarian, and dynamically allocated strategies—we position momentum within the broader landscape of portfolio construction techniques. This contextualization helps elucidate the strategic role of momentum across different investment philosophies. Second, by systematically reviewing the empirical literature on momentum effects and categorizing the various types of momentum strategies, we offer a structured and holistic account of the phenomenon. Third, by engaging with both efficient market and behavioral finance explanations, we provide critical insights into the potential drivers of momentum, thereby enhancing our understanding of its implications for asset pricing and investor behavior.

Moreover, this paper goes beyond the existing literature by providing a detailed discussion of the integration of momentum strategies with other investment strategies. We believe that this discussion is particularly relevant to practitioners and investors who are interested in implementing momentum strategies within a broader investment framework. By highlighting the potential benefits and drawbacks of combining momentum strategies with other types of investment strategies, we hope to provide practical guidance for investors looking to optimize their investment portfolios.

In sum, this paper offers a comprehensive synthesis of the momentum literature and its practical ramifications. By bridging empirical findings with theoretical insights and exploring their intersection with other investment strategies, we endeavor to advance the develop-

ment of more nuanced and effective portfolio management frameworks that can enhance risk-adjusted returns.

To provide a systematic and comprehensive overview of the existing literature, Table 1 summarizes the key studies reviewed. For each study, the table reports the main research issue, the theoretical foundations, key variables, empirical methodologies, and principle findings. Together with the subsequent sections, it serves as a concise reference that facilitates a clear assessment of the scope and contributions of the reviewed literature.

The remainder of the paper is organized as follows. Section 2 provides an overview of technical, fundamental, contrarian, and dynamic asset allocation strategies. Section 3 discusses empirical evidence supporting the existence of the momentum effect, including the relative strength approach and the cross-sectional regression framework. Section 4 surveys various momentum strategies, encompassing price momentum, 52-week high momentum, time-series momentum, persistent momentum, industry and market momentum, earnings momentum, revenue momentum, and composite momentum. Section 5 investigates the underlying sources of momentum, addressing both efficient market and behavioral explanations. Section 6 examines the role of business cycles and regime transitions in shaping momentum performance. Section 7 reviews evidence on momentum crashes and their main drivers. Section 8 analyzes short-horizon return dynamics, including the interplay between momentum, reversals, and investor clienteles. Section 9 discusses the momentum effect in cryptocurrency markets. Section 10 concludes the paper.

2 Technical analysis, fundamental analysis, and dynamic asset allocation

Before delving into momentum strategies, it is worth noting that momentum is just one of several types of investment strategies. Technical and fundamental analysis are two other popular strategies that are frequently used by investors.

Technical analysis is based on the idea that past price and volume data can be used to predict future price movements. Technical analysts use a variety of tools and techniques to identify patterns in price data, such as moving averages, trendlines, and chart patterns. They also use indicators such as the relative strength index (RSI) and moving average convergence divergence (MACD) to identify potential buy and sell signals.[1] Moreover, Lee et al. (2009, 2013) highlight the presence of a supply effect in the asset pricing process and demonstrate that the disequilibrium price adjustment process is essential for investigating asset pricing, particularly in the context of technical analysis of portfolio management.

Fundamental analysis, on the other hand, focuses on analyzing a company's financial statements and economic indicators to identify undervalued or overvalued stocks. Fundamental analysts look at factors such as revenue growth, earnings per share, and return on equity to determine the intrinsic value of a company's stock. They may also look at macroeconomic indicators such as GDP growth and interest rates to identify trends that could affect the stock market as a whole.

Contrarian strategies and dynamic asset allocation are two other investment strategies that are worth mentioning. Contrarian strategies involve going against the prevailing mar-

[1] See Lo et al. (2000), Gil (2023), and Han et al. (2023).

Table 1 Application of momentum strategies in finance research

Study	Issue	Corresponding theories, proxies, and model specifications	Main findings
Ball and Brown (1968)	Earnings momentum strategy	Net income & EPS	-Assets with positive (negative) earnings surprises tend to continue outperforming (underperforming) in the near future.
Jegadeesh and Titman (1993)	The relative strength strategy	Security's past returns in cross section	-A portfolio of high relative strength stocks significantly outperformed a portfolio of low relative strength stocks over a 3- to 12-month holding period.
Chan et al. (1996)	Composite momentum strategy	Security's past returns in cross section & earnings	-Medium-term return continuation can be explained in part by underreaction to earnings information.
Carhart (1997)	Sources of momentum effect	Security's past returns in cross section	-Common factors in stock returns and investment expenses almost completely explain persistence in equity mutual funds' mean and risk-adjusted returns.
Richards (1997)	Market momentum strategy	Security's and market past return in cross section	-Small markets are subject to larger reversals than large markets.
Barberis et al. (1998)	Sources of price continuation & long-term reversals	Conservatism/Representativeness bias	-Conservatism bias could lead to underreaction to new information, and representativeness biases result in overreaction and reversals when people overweight "representative" events.
Daniel, Hirshleifer, and Subrahmanyam (1998)	Sources of price continuation & long-term reversals	Overconfidence	-Overconfident investors initially overreact to their private information; Their confidence is enhanced upon subsequent arrivals of public signals due to biased self-attribution and leads to further overreaction to the information.
Rouwenhorst (1998)	Price momentum strategy	Security's past returns in cross section	-Momentum effects exist in several countries.
Hong and Stein (1999)	Sources of price continuation	Gradual information diffusion	-The authors model a market populated by two groups of boundedly rational agents: "news watchers" and "momentum traders." If information diffuses gradually across the population, prices underreact in the short run.
Moskowitz and Grinblatt (1999)	Industrial momentum strategy	Security's and industry past returns in cross section	-Industry momentum strategies, which buy stocks from past winning industries and sell stocks from past losing industries, appear highly profitable, even after controlling for size, book-to-market equity, and individual stock momentum.
Hong et al. (2000)	Sources of momentum effect	Gradual information diffusion	-The authors test the gradual-information-diffusion model of Hong and Stein (1999). They show that momentum strategies work better among stocks with low analyst coverage or small size. These findings are consistent with the hypothesis that firm-specific information, especially negative information, diffuses only gradually across the investing public.
Lee and Swaminathan (2000)	Sources of momentum effect	Momentum Life Cycle	-Low-volume winners persist lengthier, while high-volume losers persist lengthier. A high-volume-based momentum can a yield stronger momentum effect than JT momentum.

Table 1 (continued)

Study	Issue	Corresponding theories, proxies, and model specifications	Main findings
Piotroski (2000)	Composite momentum strategy	F-SCORE	-Firms with lower levels of financial distress (F-score) earned significantly stronger future returns than the highly distressed firms.
Jegadeesh and Titman (2001)	Sources of momentum effect & competing explanations	Underreaction/Behavioral bias/Cross-sectional returns	- The authors confirm that momentum profits persist across various subsamples and cannot be fully explained by risk-based models; instead, behavioral explanations such as delayed overreaction and investor underreaction are more consistent with the evidence.
D'Avolio (2002)	Sources of short Sales Constraints	Limit to arbitrage/Short Sales Constraints	-This paper describes the market for borrowing and lending U.S. equities, emphasizing the conditions generating and sustaining short-sale constraints. The incidence of these short-sale constraints is increasing due to the divergence of opinion among investors.
Hong et al. (2003)	Earnings momentum strategy	Analyst forecast revisions/EPS	-This paper examines the profitability of earnings momentum strategies based on analyst forecast revisions across 11 international equity markets. They find that earnings momentum is significantly profitable only in countries such as Australia, Canada, France, Germany, Hong Kong, and the UK—markets that also exhibit strong price momentum—while countries like Japan, Korea, and Singapore do not.
Pastor and Stambaugh (2003)	Sources of momentum effect	Liquidity risk factor	-A liquidity risk factor accounts for half of the profits of a momentum strategy over the same 34-year period.
Scheinkman and Xiong (2003)	Sources of momentum effect	Heterogeneous beliefs	-The authors present a continuous-time equilibrium model in which overconfidence generates disagreements among agents regarding asset fundamentals. With short sale constraints, a significant bubble component exists in asset prices even when minor differences of beliefs are sufficient to generate a trade.
Cooper et al. (2004)	Sources of momentum effect	Overconfidence	-Momentum profits are higher when the market continues in the UP state due to investors' optimism.
George and Hwang (2004)	Sources of momentum effect	Anchoring bias/52-week high ratio	-Nearness to the 52-week high dominates the forecasting power of past returns for future returns and explains a large portion of the profits from momentum investing.
Grinblatt and Han (2005)	Sources of momentum effect	Disposition effect	-A reluctance to sell losers and an eagerness to sell winners both cause the price to underreact to true fundamental news for losing stocks.
Griffin and Martin (2005)	Composite momentum strategy	Security's past returns in cross section & earnings	-Internationally, earnings momentum is distinct from price momentum, and using price and earnings momentum in conjunction produces larger economic profits. In general, momentum is useful in international portfolio management, but its implementation should be thoughtfully considered.

Table 1 (continued)

Study	Issue	Corresponding theories, proxies, and model specifications	Main findings
Mohanram (2005)	Composite momentum strategy	G-SCORE	-High G-SCORE firms have a greater market reaction and analyst forecast surprises with respect to future earnings announcements. A long–short strategy based on GSCORE earns significant excess returns, though most of the returns come from the short side.
Jiang et al. (2005)	Sources of price & earnings momentum effect	Overconfidence	-Using several different proxies for IU, the authors show that, on average, high-uncertainty firms earn lower future returns, and price and earnings momentum effects are much stronger among high-uncertainty firms.
Ali and Trombley (2006)	Sources of momentum effect	Limit to arbitrage/Short Sales Constraints	-The magnitude of momentum returns for the period 1984 to 2001 is positively related to short sales constraints, and loser stocks rather than winner stocks drive this result.
Chordia and Shivakumar (2006)	Composite momentum strategy	Earnings surprises	-Earnings momentum and price momentum are related, and price momentum is captured by the systematic component of earnings momentum. The predictive power of past returns is subsumed by a zero-investment portfolio that is long on stocks with high earnings surprises and short on stocks with low earnings surprises.
Jegadeesh and Livnat (2006a)	Composite momentum strategy	Several fundamental metrics	-Strategy combines several fundamental metrics, including earnings, book-to-market ratio, and cash flow, to identify undervalued companies with strong momentum.
Jegadeesh and Livnat (2006b)	Revenue momentum strategy	Revenue surprises	-Assets with positive earnings surprises tend to continue performing well in the near future. Post-earnings-announcement drift was found to be stronger when the revenue surprise was in the same direction as the earnings surprise.
Zhang (2006)	Sources of momentum effect	Overconfidence	-Information uncertainty is higher, and price continuation would be more pronounced.
Gutierrez and Prinsky (2007)	Sources of momentum & long-term reversals	Institutional trading	-The results show that relative-return momentum and abnormal-return momentum are consequences of agency issues in the money management industry and provide empirical support for this economic rationale of momentum in returns. Incentives induce institutions to chase relative returns and to underreact to firm-specific abnormal returns.
Hong and Stein (2007)	Sources of investor disagreement	Heterogeneous beliefs	-Disagreement among investors could result from investors' differential priors (i.e., heterogeneous beliefs) or data assessments (i.e., the "limited attention" or "gradual information flow"). These cause a large portion of trading in the market.
Sagi and Seasholes (2007)	Composite momentum strategy	Security's and industry past returns in cross section and fundamental variables	-The combination of industry momentum, fundamental momentum, and price momentum. They found that their strategy generates higher returns than traditional momentum strategies.

Table 1 (continued)

Study	Issue	Corresponding theories, proxies, and model specifications	Main findings
Arena et al. (2008)	Sources of momentum effect	Limit to arbitrage/ Idiosyncratic risks	-Higher idiosyncratic risk impedes the correction to mispricing and thus leads to stronger momentum effect.
Kaniel et al. (2008)	Retail contrarian trading & short-horizon patterns	Uses net individual investor trading; interprets individuals as contrarian liquidity providers	-Individuals buy after declines/sell after increases; connects individual trading to short-horizon return patterns, supporting "retail-driven reversal/ liquidity provision" channel.
Liu and Zhang (2008)	Sources of momentum effect	Macroeconomic risk/ CRR(1986) factors	-The momentum effect can be explained more than half by macroeconomic risk factors, such as the growth rate of industrial production (MP).
Asem (2009)	Composite momentum strategy	Security's past returns in cross section & dividend	-Buying winners with increased dividends and shorting losers with decreased dividends enhances momentum profits.
Verardo (2009)	Sources of momentum effect	Heterogeneous beliefs/Dispersion in analysts' forecasts	-The results show that momentum profits are significantly larger for portfolios characterized by higher heterogeneity of beliefs.
Asem and Tian (2010)	Sources of momentum effect	Overconfidence	-Transitions in the market state play an important role in driving momentum profits. Investor overconfidence is strengthened when the market continues in the same direction.
Chui et al. (2010)	Sources of momentum effect	Individualism	-Individualistic countries generate more significant momentum profits than collectivistic countries.
Baker et al. (2012)	Sources of momentum effect	Investor sentiment	- The authors find that both global and local sentiment are contrarian predictors of the time-series of cross-sectional returns within markets: When sentiment is high, future returns are low on relatively difficult to arbitrage and difficult to value stocks.
Moskowitz et al. (2012)	Time-series momentum strategy	Security's own past return	-The authors find persistence in returns for one to 12 months that partially reverses over longer horizons in equity index, currency, commodity, and bond futures for each of the 58 liquid instruments, consistent with sentiment theories of initial underreaction and delayed overreaction.
Nagel (2012)	Liquidity provision view of short-term reversals	Interprets reversal-strategy returns as returns to liquidity provision; time-varying with market stress (e.g., VIX)	-Short-term reversal profits behave like liquidity provision returns and become strongly time-varying in stressed period
Novy-Marx (2012)	Sources of momentum effect	Security's past returns in cross section	-Momentum is primarily driven by firms' performance 12 to seven months prior to portfolio formation.
Stambaugh et al. (2012)	Sources of momentum effect	Investor sentiment	-Overpricing is the primary form of mispricing due to short selling costs, and the momentum should be relatively more prevalent when sentiment is high.
Antoniou et al. (2013)	Sources of momentum effect	Investor sentiment	-Negative (positive) information about stocks, when sentiment is optimistic (pessimistic), will conflict with investors' prior beliefs and thus cause cognitive dissonance.

Table 1 (continued)

Study	Issue	Corresponding theories, proxies, and model specifications	Main findings
Asness et al. (2013)	Composite momentum strategy	Security's past returns in cross section & BM ratio	-Value and momentum returns correlate more strongly across asset classes than passive exposures to the asset classes, but value and momentum are negatively correlated with each other, both within and across asset classes.
Bandarchuk and Hilscher (2013)	Price momentum	Security's past returns in cross section & characteristics	-We show that such strategies enhance momentum profits simply by trading in stocks with more extreme past returns.
Chen et al. (2014)	Composite momentum strategy	Security's past returns in cross section & revenue surprises & earnings surprises	-Revenue surprises, earnings surprises, and prior returns each carry some exclusive unpriced information content.
Barroso and Santa-Clara (2015)	Momentum crash and time-varying risk	Volatility-managed momentum & scaling momentum exposure by inverse realized volatility	-Momentum risk is highly predictable using realized volatility. -Volatility-managed momentum virtually eliminates momentum crashes and nearly doubles the Sharpe ratio relative to the standard momentum strategy.
Daniel and Hirshleifer (2015)	Sources of price continuation & long-term reversals	Overconfidence	-The authors describe how overconfident investors respond to public and private information and generate excessive trading, and thus induce predictable return patterns in the markets.
Byun et al. (2016)	Sources of momentum effect	Overconfidence	-The authors show that a measure of continuing overreaction based on the weighted average of signed volume is a better predictor of future returns than past returns that subsumes the price momentum effect.
Chen et al. (2016)	Composite momentum strategy	Security's past returns in cross section & F-SCORE & G-SCORE	-The combined strategy includes stocks with a larger information content from the scores that the market cannot reflect in time, and outperforms momentum.
Daniel and Moskowitz (2016)	Sources of momentum effect	Time-varying beta/Market rebound/Dynamic weighting	-Momentum crashes arise due to conditional beta shifts—particularly when momentum portfolios are short high-beta losers during sharp market rebounds—exhibiting option-like payoffs that result in large losses after bear markets; a dynamic weighting strategy that scales exposure based on forecasted Sharpe ratios significantly reduces crash risk and outperforms static and constant-volatility strategies across asset classes.
Moreira and Muir (2017)	Risk-managed momentum and factor timing	Volatility-managed factors & scaling factor exposures by aggregate market volatility	-Volatility-managed momentum achieves substantially higher Sharpe ratios and lower drawdowns. -Momentum crashes are closely linked to periods of elevated market volatility, highlighting the importance of volatility timing in factor investing.
Chen et al. (2018)	Persistent momentum strategy	Security's past returns in cross section	-A persistent momentum strategy by buying persistent winners and selling persistent losers can generate stronger momentum returns than JT's momentum.

Table 1 (continued)

Study	Issue	Corresponding theories, proxies, and model specifications	Main findings
Grobys et al. (2018)	Risk-managed industry momentum and crash robustness	Industry-level momentum strategies; volatility-based risk management; alternative variance forecast horizons	-Finds that volatility-based risk management remains effective in mitigating momentum crashes at the industry level. -Shows robustness of crash mitigation even when momentum exhibits limited time-varying beta.
Gulen and Petkova (2018)	Absolute Strength strategy	Security's past returns in cross section & own past return	-Stocks that have significantly increased in value in the recent past (absolute strength winners) continue to gain, and stocks that have significantly decreased in value (absolute strength losers) continue to lose in the near future. Absolute strength momentum generates large and significant risk-adjusted returns and outperforms JT's momentum strategies.
Li et al. (2018)	Sources of momentum effect	Implied volatility/ Option prices/Market volatility	-In the Chinese market, option-implied volatility and stock returns are strongly linked; during high market volatility periods, changes in implied volatility predict the strength of momentum profits, indicating that derivatives market information plays a crucial role in momentum dynamics.
Grobys and Sapkota (2019)	Short-horizon cryptocurrency momentum	Weekly winner–loser portfolios based on past returns	-Cryptocurrencies exhibit significant momentum at short horizons. -Momentum profits dissipate quickly as holding periods lengthen.
Li et al. (2019)	Size effects and return reversals	Broad cryptocurrency universe; size-sorted portfolios	-Small and illiquid cryptocurrencies exhibit strong short-term reversals. -Asset coverage critically affects return predictability.
Yang and Zhang (2019)	Extreme absolute strength strategy	Security's past returns in cross section & own past return	-The removal of extreme absolute strength reduces the risk of the momentum portfolios and improves their performances.
Atilgan et al. (2020)	Sources of momentum effect	Gradual information diffusion	-This paper documents a significantly negative cross-sectional relation between left-tail risk and future returns on individual stocks trading in the US and international countries.
Shen et al. (2020)	Factor pricing and reversals	Three-factor model for cryptocurrencies	-Return predictability varies across asset universes and horizons. -Evidence of short-term reversals in broad samples.
Tzouvanas et al. (2020)	Momentum and diversification benefits	Short-term momentum strategies in cryptocurrencies	-Short-horizon momentum strategies generate significant abnormal returns. -Momentum offers diversification benefits relative to traditional assets.
Kelly et al. (2021)	Sources of momentum effect	Conditional asset pricing model	-Conditional risk exposure that depends on observable firm characteristics can explain a sizable fraction of momentum and long-term reversal returns.
Lin et al. (2021)	Time-dependent MAX strategy	Historical distribution of MAX for a security	-A time-dependent MAX strategy by buying stocks with the recent MAX that fall below the historical distribution's 10th percentile and selling those above the 90th percentile.

Table 1 (continued)

Study	Issue	Corresponding theories, proxies, and model specifications	Main findings
Liu and Tsyvinski (2021)	Time-series momentum in cryptocurrencies	Time-series momentum effect in crypto; return predictability framework	-Documents a strong time-series momentum effect in cryptocurrencies (as part of broader risk/return characterization). (OUP Academic)
Zaremba et al. (2021)	Reconciliation of momentum vs. reversal evidence	Comparative review across samples and horizons	-Disagreements in the literature largely stem from asset coverage and horizon choice. | Momentum is more likely in narrow, liquid samples; reversals dominate in broad universes.
Chui et al. (2022)	Investor clienteles as a natural experiment (China A vs. B shares)	Same-firm share-class design isolates clientele effects; A shares more retail, B shares more institutional; links to earnings drift and return predictability	-Only B shares show momentum (and earnings drift) and only A shares show monthly reversals -investor composition can flip the sign of short-horizon return dependence even within the same firm. (OUP Academic)
Dierkes and Krupski (2022)	Identification and isolation of momentum crash states	Ex-ante crash indicators combining market states, momentum volatility, and momentum beta; crash-conditioned trading strategies	-Shows that momentum crashes systematically occur in rebounding bear markets with high volatility and negative momentum betas. -Crash indicators effectively isolate crash periods and further improve risk-adjusted performance beyond existing volatility-managed approaches.
Ehsani and Linnainmaa (2022)	Sources of momentum effect	Factor momentum	-Factor momentum transmits into the cross-sectional stock returns through variation in stocks' factor loadings, which explains JT's and other momentum.
Liu et al. (2022)	Momentum as a systematic risk factor	Cryptocurrency factor-pricing model with momentum factor	-Short-horizon momentum earns a significant risk premium. -Momentum is a systematic component of cryptocurrency returns.
Medhat and Schmeling (2022)	Short-term reversal and short-horizon momentum	Past 1-month return/ Turnover/Trading intensity as clientele proxy)	-High-turnover stocks exhibit short-term momentum, while low-turnover stocks exhibit short-term reversal -short-horizon return dependence depends on investor segment and trading environment. (OUP Academic)
Chen et al. (2023)	Persistent momentum strategy	Security's past returns in cross section	-Persistent momentum effect outperforms JT's momentum in either momentum or non-momentum market.
Hanauer and Windmüller (2023)	Enhanced momentum strategies	Volatility-scaled momentum/Semi-volatility scaling/ Dynamic scaling	-Three enhanced momentum strategies (constant volatility-scaled, semi-volatility-scaled, and dynamic-scaled) significantly increase Sharpe ratios and reduce momentum crashes.
Goulding et al. (2023)	Sources of momentum effect	Business cycle/ Turning point	-Momentum strategies suffer the most at market turning points, particularly during Bear states that cluster in early-stage recessions, while combining slow and fast time-series momentum signals significantly improves timing accuracy and yields higher Sharpe ratios across regimes.

Table 1 (continued)

Study	Issue	Corresponding theories, proxies, and model specifications	Main findings
Abhyankar et al. (2024)	Sources of momentum effect	Short-sale constraints/ Option trading	-The role of option markets in mitigating short-sale constraints that hinder momentum arbitrage. They show that the decline in momentum profits over the past two decades is largely attributable to the expansion of stock options trading, which serves as a substitute for costly or restricted short selling.
Büsing et al. (2024)	Sources of momentum effect	Pre-peak return & momentum decomposition	- Momentum profits are largely driven by the pre-peak return component, consistent with underreaction and robust to crashes and market conditions.
Chen et al. (2024)	Sources of momentum effect	Limit to arbitrage	- Momentum is evident for stocks with both a relatively high price-to-52-week-high ratio and turnover.
Fu et al. (2024)	Composite momentum strategy	Free cash flow	-A strategy of buying past winners with high free cash flows and shorting past losers with low free cash flows significantly outperforms the traditional momentum trading strategy.
Ma et al. (2024)	Sources of momentum effect	Factor momentum	- Factor momentum subsumes stock momentum, high-priced stock momentum, and industry momentum in China.
Cakici et al. (2025)	Sources of momentum effect	Factor momentum	- The authors demonstrate that across 51 international equity markets, price momentum more often explains factor momentum than vice versa, challenging the view that factor momentum subsumes price momentum and reaffirming momentum as a distinct, pervasive, and robust market anomaly.
Du et al. (2025)	Sources of momentum effect	Limit to arbitrage	- Price momentum is substantially stronger in high-nominal-price stocks, where round-lot trading rules limit retail participation, implying that momentum is more detectable when noise trading is attenuated.
Fieberg et al. (2025)	Trend/momentum factor in crypto cross-section	Cross-sectional "trend factor" for cryptocurrency returns	-Proposes and tests a trend-based factor for the cross-section of cryptocurrency returns.
Goyal et al. (2025)	Sources of momentum effect	Gradual information diffusion	- Using firm-level data from both developed and emerging markets, they find that momentum is strongest when information arrives gradually rather than in large chunks
Hou et al. (2025)	Sources of momentum effect	Investor attention	- Investor attention drives momentum through a dual mechanism: inattention causes underreaction to earnings news, while heightened attention amplifies behavioral biases fueling price momentum.
Hsieh et al. (2025)	Regime-dependent momentum	Transition-based market state framework	-Momentum arises only in persistent up-market regimes. | Momentum is path-dependent and sensitive to regime continuity.

Table 1 (continued)

Study	Issue	Corresponding theories, proxies, and model specifications	Main findings
Jegadeesh et al. (2025)	Short-term reversal and longer-term momentum transition	Multiperiod model with short-/long-horizon noise traders + active investors who underreact to information they do not originate; generates testable predictions on the reversal-to-momentum transition	Provides a theoretical foundation: reversals arise from short-horizon noise/price pressure; momentum arises from trading on information over time

ket trends and investing in assets that are currently out of favor. Dynamic asset allocation involves adjusting a portfolio's asset allocation based on changing market conditions.

3 The existence of the momentum effect

One of the most extensively studied investment strategies is momentum investing. The momentum effect refers to the observation that assets with strong past performance tend to continue to perform well in the future. This effect has been observed in a wide range of financial markets, including stocks, bonds, currencies, and commodities.

3.1 The relative strength strategy

The relative strength strategy is one of the earliest and most well-known momentum strategies. It involves ranking assets based on their past performance relative to other assets in the same market or sector. Assets that have outperformed their peers are considered to have high relative strength and are more likely to continue to perform well in the future. Jegadeesh and Titman (1993) are among the first researchers to document the momentum effect in U.S. stock returns using the relative strength strategy. They construct winner and loser portfolios by the following procedure: at the end of each month, they rank sample stocks into deciles according to their prior J-month returns ($J=3$, 6, 9, and 12) and group the stocks into 10 equally weighted portfolios. Stocks in the top decile portfolio are called 'winners,' and those in the bottom decile are called 'losers.'

Forming portfolios based on past three- to 12-month returns, they show that past winners, on average, continue to outperform past losers over the next three to 12 months. The significant "momentum" phenomenon, therefore, has attracted a substantial body of theoretical and empirical research aimed at understanding this pattern in stock markets. Rouwenhorst (1998) and Chui et al. (2010) also confirmed the momentum effect in international stock markets. They find that, outside the U.S. market, momentum strategies are profitable in several countries they studied, including Australia, Canada, France, Germany, Hong Kong, Italy, Japan, the Netherlands, Singapore, Spain, Sweden, the United Kingdom, and others.

3.2 The cross-sectional regression method

George and Hwang (2004) propose an alternative method to measure the momentum effect. They consider a Fama-MacBeth cross-sectional regression with dummy variables. Unlike common Fama–MacBeth regression applications, where the explanatory variables of interest take the form of continuous measures, they use dummy variables to capture the "net" return of the portfolio based on the coefficient of the dummy variable. The original model can be presented as follows:

$$R_{it} = b_{0jt} + b_{Hjt} JH_{it}\,(j) + b_{Ljt} JL_{it}\,(j) + \varepsilon_{it,} i = 1, \ldots, N, \tag{1}$$

where $JH_{i,\,t-j}$ ($JL_{i,\,t-j}$) equals one if stock i's past performance over the 6-month period ($t - j - 6$, $t - j$) is in the top (bottom) 30% when measured by JT's performance criterion, and is zero otherwise. Because the high- and low-return portfolios are formed based on the past performance of the top and bottom 30% stocks, $\widehat{b}_{0jt}$ is the average return of the "medium" portfolio which is composed of the other 70% stocks in the sample. Here, $\widehat{b}_{Hjt}$ ($\widehat{b}_{Ljt}$) is the *incremental* return on the high-return (low-return) portfolio over the medium portfolio. At time t, the total return on the high-return (low-return) portfolio formed at time $t-j$ is $\widehat{b}_{0jt} + \widehat{b}_{Hjt}$ ($\widehat{b}_{0jt} + \widehat{b}_{Ljt}$). Clearly, $b_{H}jt$ ($bLjt$) captures the equal-weighted return of winner (loser) stocks formed at $t-j$ (i.e., the jth -month return following formation at time $t-j$).

Although the regression setting fully conforms to momentum portfolio, the returns on the winner and loser portfolios may not best reflect the true performance of the trading strategies because there are microstructural problems because of nonsynchronous trading and the bid-ask bounce. Therefore, the regression model can be adjusted as follows:

$$R_{i,t} = b_{0j,t} + b_{1j,t} R_{i,t-1} + b_{2j,t} Size_{i,t-1} + b_{Hj,t} JH_{i,t}\,(j) + b_{Lj,t} JL_{i,t}\,(j) + \varepsilon_{i,t,} \tag{2}$$

where $R_{i,\,t-1}$ and $Size_{i,\,t-1}$ are the return and market capitalization of stock i at time $t-1$, respectively, which are included to control for the size and microstructure effects because of the bid-ask bounce. The coefficient estimate of $b_{H,\,jt}$ ($b_{L,\,jt}$) the month t return to a zero investment portfolio that is long JT winner (loser) stocks but that has also hedged out all other effects. Consistent with Jegadeesh and Titman (1993), George and Hwang (2004) show evidence that the coefficient estimate of $b_{H,\,jt}$ ($b_{L,\,jt}$) is positively (negatively) significant, and the magnitude of the return spreads between the two coefficients are also significant. The authors thus suggest that the price momentum effect exists with consideration of microstructural problems.

4 Momentum strategies

Momentum strategies involve identifying and investing in assets that have exhibited strong past performance. These strategies are based on the assumption that assets that have performed well in the past are likely to continue performing well in the future. Momentum strategies can be based on technical or fundamental information or a combination of both.

4.1 Technical information

Technical information plays a pivotal role in identifying assets with strong momentum. To bridge the gap between technical analysis and quantitative finance, Lo et al. (2000) propose a systematic and scientific approach to integrate computational algorithms and statistical inferences. They suggest that technical analysis can complement and enhance the investment process. Han et al. (2023) provide a comprehensive review of the forecasting capabilities of technical indicators in stock market trends. Gil (2023) underscores the value of technical analysis in refining investment decisions. These insights lay the foundation for deeper dives into specific technical strategies that vary in their approach and efficacy within the broader context of momentum investing. Notably, the momentum literature frequently highlights technical indicators, including the 52-week high momentum, time-series momentum, persistent momentum, and industry and market momentum.

4.1.1 Price momentum

Taylor (1982) challenges the random walk hypothesis by highlighting the existence of price trends, suggesting that price patterns are not purely random but possess discernible trends. When Jegadeesh and Titman (1993) first document the effectiveness of price momentum strategies, it represents a seminal contribution in recognizing the power of historical price data to predict future price movements. Their findings underscore the potential of technical analysis, where stock price return can be a crucial determinant in identifying assets with strong momentum. Rouwenhorst (1998) and Chui et al. (2010) reinforce this finding by showing that price momentum is a global phenomenon. In addition, Cooper et al. (2004) find that price momentum strategies can generate excess returns in bullish markets. Asem and Tian (2010) further show that the momentum effect is more pronounced when states continue in the same state. These studies confirm the idea that stock price return is not just random noise but contains information about collective sentiment and expectations of market participants.

4.1.2 52-week high momentum

George and Hwang (2004) propose an alternative 52-week high momentum strategy based on the nearness of a stock's current price to its highest price achieved during the past 52 weeks, which involves buying stocks whose 52-week high ratio is ranked at the top decile and selling stocks whose 52-week high ratio is ranked at the bottom decile. The measure is defined as below:

$$52 - \text{Week High} = \frac{\text{Current price}}{52 - \text{week high price}} \tag{3}$$

The higher values of the 52-week high ratio indicate that the current price of a stock is closer to its 52-week high price. The highest possible value of the 52-week high ratio is 1, which occurs when the end-of-month price is the 52-week high price. The authors show that when stocks are sorted into portfolios based on the ratio of their current price to their 52-week

high price, the stocks with the highest ratios significantly outperform the firms with the lowest ratios over the subsequent 6- to 12-month period, particularly for small and value stocks.

Furthermore, the authors find that this specific strategy dominates and outperforms the traditional momentum strategy. They attribute this finding to an ‘‘adjustment and anchoring’’ bias (Kahneman et al. 1982) that causes investors to underreact to positive (negative) information about stocks whose prices are near (far from) their 52-week high prices. They suggest that if traders consider available price information, the 52-week high price, they would underreact to new information when the current stock price close to the 52-week high price. People use the 52-week high as an “anchor” when assessing the increment in stock value implied by new information. Nearness to the 52-week high dominates the forecasting power of past returns for future returns and explains a large portion of the profits from momentum investing.

4.1.3 Time-series momentum

Chordia and Shivakumar (2002) argue that the time variation in momentum returns is dependent on the state of the economy. Their paper shows that the profits to momentum strategies are explained by common macroeconomic variables that are related to the business cycle. Momentum profits are positive only during expansionary periods, while during recessions, the momentum strategy returns are negative. Using a set of lagged macroeconomic variables to predict one-month ahead returns, including dividend yield, default spread, yield on three-month T-bills, and term structure spread, they show that the momentum portfolios formed on the basis of past returns vary systematically in their sensitivity to these macroeconomic variables. Moreover, they find that industry-based momentum returns are also captured by macroeconomic variables.

Moskowitz et al. (2012) show that the ‘‘time-series momentum’’ strategy can generate significant positive returns in equity index, currency, commodity, and bond futures, even after accounting for transaction costs. Unlike price momentum strategy, which focuses on the relative performance of securities in the cross-section, time-series momentum strategy focuses purely on a security’s own past return, which involves buying stocks with positive excess returns over the past k months and selling stocks with negative excess returns over the same period, holding for h months. They find persistence in returns for one to 12 months that partially reverses over longer horizons, consistent with sentiment theories of initial underreaction and delayed overreaction. Lin et al. (2021) introduce the time-dependent MAX strategy, in which MAX is defined as the most recent maximum daily return. The strategy involves the acquisition of stocks with recent MAX values lower than the historical distribution’s 10th percentile and the short-selling of those exceeding the 90th percentile. Their findings suggest that such a strategy’s profitability stems from investors betting on stocks with notably recent performance against their historical records.

Among the vast literature on the sources of momentum, two lines of research have proposed distinct explanations for momentum based on extreme past performances of stocks. Bandarchuk and Hilscher (2013) show that momentum profits can be enhanced by trading in stocks with extreme past returns. Once this effect is controlled, the explanatory power of several firm characteristics disappears almost entirely. Gulen and Petkova (2018) propose that the past performance of a stock should be evaluated compared with the entire historical distribution of returns rather than just compared with contemporary benchmark returns in the

cross-section. Yang and Zhang (2019), in contrast, propose that stocks with extreme absolute strength, such as those experiencing substantial positive or negative returns recently, have extraordinarily high volatility and thus lose their momentum. Removing these stocks from the winner and loser portfolios of Jegadeesh and Titman's (1993) price momentum strategy is able to enhance momentum profitability.

4.1.4 Persistent momentum

Among the extensive literature on sources of momentum, one strand focuses on the connection between the persistency of past returns and future price movements. Watkins (2003, 2006) suggest that, unlike the magnitude of returns, persistency is informative about how path-dependent stock prices are. The predictive power of return persistency for future price movements is distinct from that of past return magnitude. Based on this idea, Chen et al. (2018) identify persistent momentum as a new momentum strategy that involves buying *persistent winners* and selling *persistent losers*. By their definition, a *persistent winner* is a winner stock in which a winner (loser) stock is consecutively ranked in the top (bottom) decile portfolio after the formation month, while a *persistent loser* is a loser stock in which a winner (loser) stock is consecutively ranked in the top (bottom) decile portfolio after the formation month. Winners and losers with a *zero duration* mean that the stock immediately falls out of the winner (loser) portfolio after it is initially identified as a winner (loser); such stocks are called *nonpersistent winners* (*losers*).

They demonstrate that the observation that a change of only one-sixth of the formation-period return will knock nearly 48% of winner and loser stocks out of their respective groups confirms the importance of the role of the subsequent monthly returns. Winners and losers that immediately fall out of their respective groups in the month following formation will result in a monthly loss of more than 17% for a momentum strategy constructed on such stocks. They found that this strategy can generate much more significant positive returns than the price momentum strategy, particularly during market crises.

Chen et al. (2023) extend the sample to international markets, including markets that exhibit insignificant price momentum effects. The results show that for markets exhibiting an insignificant momentum effect, the persistent momentum strategy can generate significant momentum profitability compared with the price momentum strategy. This indicates that lengthier persistence of past performance can help investors develop a trading strategy that enhances momentum profitability. In addition, for markets with a significant momentum effect, the persistent momentum strategy can still yield higher momentum returns than the price momentum strategy. Thus, they conclude that the persistent momentum strategy can perform well for both insignificant and significant momentum markets.

4.1.5 Industry and market momentum

Some studies suggest that market-wide information can help to explain the price momentum effect on individual stocks. Richards (1997) examines possible explanations for price momentum in the national stock market indices of 16 countries. This paper finds that a country-based momentum strategy, which involves buying assets that have shown strong performance during bull markets and selling them during bear markets, can generate strong

return continuation. This indicates that the existence of the momentum effect on individual stocks is probably attributed to the persistence of country-specific market performance.

Moskowitz and Grinblatt (1999) study the industry-based momentum. This paper finds that industry momentum strategies, which buy stocks from past winning industries and sell stocks from past losing industries, appear highly profitable, even after controlling for individual stock momentum and fundamental variables. By contrast, individual stock momentum strategies, which buy past winning stocks and sell past losing stocks, are significantly less profitable once controlling for industry momentum.

4.2 Fundamental information

Fundamental information involves analyzing financial statements to identify assets with strong momentum. Some of the most widely studied fundamental indicators include earnings momentum, revenue momentum, and composite momentum.

4.2.1 Earnings momentum

Firm disclosures such as earnings announcements have been given as an example of public information since the earliest days of pricing theory. For just as long, current studies have accumulated evidence that news from firm disclosures is not impounded in prices as quickly and efficiently as the efficient market hypothesis predicts. These findings motivate a central question in the disclosure processing literature: are firms' disclosures actually public information?

Ball and Brown (1968) are the first to document the importance of earnings announcements in return continuation. The authors start by acknowledging the fundamental role of financial statements in providing information to investors and market participants. They specifically focus on earnings as a key metric used by investors to evaluate a firm's financial health. They find that assets with positive earnings surprises tend to continue performing well in the near future.

The paper also highlights the importance of the timeliness of earnings information. The authors find that earnings announcements have the most significant impact on stock prices when they are timely and accurate. In contrast, delayed or unreliable earnings information leads to a diminished market reaction, indicating that timely disclosure of earnings is crucial for efficient capital markets. Moreover, the authors explore the persistence of earnings surprises. They find that firms that consistently report positive earnings surprises experience a positive market reaction over an extended period. This suggests that consistently positive earnings surprises may signal a firm's ability to generate future earnings growth, which positively impacts stock prices.

Hong et al. (2003) examine the profitability of earnings momentum strategies based on analyst forecast revisions across 11 international equity markets. They find that earnings momentum is significantly profitable only in countries such as Australia, Canada, France, Germany, Hong Kong, and the UK—markets that also exhibit strong price momentum—while countries like Japan, Korea, and Singapore do not. Their findings reveal a strong one-to-one relationship between earnings and price momentum, supporting behavioral theories of investor underreaction (e.g., Barberis et al. 1998; Daniel et al. 1998). They further suggest

that institutional features, particularly investor protection and information dissemination mechanisms, play a key role in shaping the presence and magnitude of momentum effects.

Myers et al. (2007) explore the prevalence and implications of firms consistently reporting non-decreasing quarterly EPS over extended periods. They identify 746 firms that achieved at least 20 consecutive quarters of EPS increases—a frequency significantly above what randomness would predict—interpreted as indicative of earnings management. These firms exhibit striking abnormal returns averaging over 20% annually during the early phase of the streak, with gains sharply reversing once the streaks break. The authors argue this pattern generates strong incentives for managers to sustain such earnings momentum. Through a series of empirical tests, they provide evidence that managers employ a range of earnings management practices, including accrual manipulation, tax planning, and share repurchases, to prolong these EPS growth patterns.

A recent paper by Hou et al. (2025) further finds that earnings momentum profits are significantly stronger among low-attention stocks, suggesting that limited investor attention leads to underreaction to earnings news. Using proxies such as Bloomberg terminal usage and media coverage, they show that heightened attention accelerates information incorporation and thus diminishes post-earnings drift. This supports the view that earnings momentum arises from investor inattention rather than behavioral overreaction.

Overall, the study of Ball and Brown (1968) remains a classic in accounting research, providing valuable insights into the role of accounting earnings in stock market valuations and highlighting the importance of high-quality financial reporting.

4.2.2 Revenue momentum

As mentioned in Ball and Brown (1968), one of the most important fundamental firm information for investors is undoubtedly firm earnings, which summarize firm performance. Building on this, Lee and Zumwalt (1981) highlight the associations between various accounting profitability measures and security returns, emphasizing the potential predictive power of these measures in momentum investing.

Jegadeesh and Livnat (2006a) investigate the relationship between revenue surprises (unexpected changes in a company's revenue) and subsequent stock returns. Revenue surprises are considered an important aspect of fundamental analysis, as they can reflect unexpected changes in a firm's business prospects. This paper analyzes a large dataset of U.S. stocks and focuses on the market reaction to revenue surprises. They find that positive revenue surprises tend to generate significant positive abnormal returns, while negative surprises lead to negative abnormal returns. These abnormal returns persist over the following months, suggesting that revenue surprises contain valuable information for investors. The evidence also shows that the market's reaction to revenue surprises is more pronounced during periods of high market uncertainty and for firms operating in industries with high growth potential. The paper's findings highlight the importance of revenue surprises in driving stock returns and suggest that investors can benefit from incorporating revenue surprise information into their investment decisions.

Jegadeesh and Livnat (2006b), however, point out that another important reference for investors regarding the persistence of firm earnings is offered by firm revenue information. They examine the relationship between revenue surprises and contemporary and future stock returns. Assets with positive revenue surprises continue performing well in the near

future. After controlling for earnings surprises, there still exists significant abnormal returns in the post-announcement period for stocks that have large revenue surprises. This probably implies that, although analysts revise their forecasts of future earnings in response to revenue surprises, they are slow to incorporate fully the information in revenue surprises.

4.2.3 Composite momentum

Beyond the single fundamental factor that creates return continuation, composite momentum strategies that involve combining multiple fundamental indicators may have stronger momentum than others. Prior researchers have studied this approach extensively and identified several composite momentum strategies that can generate significant positive returns. Moreover, as Guerard and Mark (2020) suggest, any composite model strategy complements fundamental data (e.g. earnings, book value, cash flow, sales, dividends, liquidity) and price momentum strategies in a composite model for stock selection can earn excess returns greater in international stocks than in US stocks.

Piotroski (2000) focuses on value investing and proposes a financial scorecard approach to identify potentially profitable investment opportunities. This paper develops a scoring system (F-Score) that evaluates various financial metrics derived from a company's financial statements, which include profitability, liquidity, leverage, operational efficiency, and other fundamental factors. A higher score indicates a stronger financial position. The author then tests the effectiveness of this scoring system by applying it to a broad sample of low-price-to-book-value firms. The study reveals that by investing in companies with high scores, investors can achieve significant positive returns. Furthermore, these returns are not attributable to higher systematic risk, suggesting that the scorecard approach identifies undervalued companies.

Mohanram (2005) similarly conducts a scoring system and constructs the G-Score (Score of growth stock), including earnings, cash flows, earnings stability, growth stability and intensity of R&D, capital expenditure, and advertising. The evidence shows that financial ratios, particularly accruals and cash flows, have predictive power in identifying future winners and losers. Low B/M stocks with higher cash flows and lower accruals tend to outperform those with weaker cash flows and higher accruals. Furthermore, stocks with high values of G-Score have greater market reactions and analyst forecast surprises with respect to future earnings announcements than those with low G-Score. A long–short strategy based on G-Score earns significant excess returns, though most of the returns come from the short side. The findings suggest that financial statement analysis can enhance investment strategies by improving stock selection within the low B/M universe.

Overall, these studies suggest that composite momentum strategies can be effective in generating significant positive returns. By combining several fundamental indicators, investors can identify companies with strong momentum and potentially outperform the market.

4.3 Combined information

In addition to technical and fundamental information, investors can also use a combination of both to develop composite momentum strategies. Composite momentum strategies can generate higher returns than strategies based solely on technical or fundamental information.

Chan et al. (1996) examine whether the predictability of past returns on future returns is due to the market's underreaction to past earnings news. They show that both price momentum and earnings momentum generate significant positive returns, even after controlling for the other. The patterns are not related to market risk, size, and book–to–market effects. However, they find that price momentum can be explained in part by underreaction to earnings information. The results suggest a market that responds only gradually to new information.

Griffin et al. (2005) extend prior studies in 40 markets for price momentum and in 34 markets for earnings momentum by analyzing the interaction between price and earnings momentum. They find that medium-term return continuation can be explained in part by underreaction to earnings information, and the combination of earnings and price momentum is more robust than either strategy alone.

Chordia and Shivakumar (2006) explore the relationship between earnings momentum and price momentum in stock returns. The authors propose that the combination of earnings momentum and price momentum can provide valuable insights for investors. They argue that earnings news can influence investor expectations and subsequent stock price movements. The evidence suggests that stocks with high earnings momentum combined with positive price momentum tend to exhibit stronger future performance. Conversely, stocks with negative earnings momentum and negative price momentum tend to underperform. The study highlights the importance of considering both earnings momentum and price momentum when formulating investment strategies. By incorporating these factors, investors can potentially exploit market inefficiencies and generate superior returns.

Sagi and Seasholes (2007) develop a composite momentum strategy that combines industry momentum, fundamental momentum, and price momentum. They confirm that a combination of industry momentum, fundamental momentum, and price momentum exists and find that those strategies generates higher returns than traditional momentum strategies.

Asem (2009) explores the relationship between dividends and price momentum in financial markets. The findings suggest a positive and statistically significant relationship between dividends and price momentum. Specifically, the research reveals that firms paying higher dividends are more likely to exhibit positive price momentum. This implies that companies with a dividend policy that emphasizes higher payouts tend to experience upward price trends in their stocks.

Asness et al. (2013) show that combining price momentum, earnings momentum, and value momentum can generate even higher returns. This paper explores the joint performance of value and momentum strategies across multiple asset classes, including equities, bonds, commodities, and currencies. The evidence shows that value and momentum strategies not only work well individually within each asset class but also exhibit positive performance when combined across asset classes. The results indicate that value and momentum are pervasive investment factors that can generate excess returns in a wide range of markets. The paper has significant implications for asset allocation and portfolio construction.

Chen et al. (2014) develop a composite momentum strategy that combines technical, fundamental, and macroeconomic variables and investigate whether revenue momentum drives or simply follows the patterns observed in earnings momentum and price momentum. The evidence shows that there is no dominating momentum strategy among the revenue, earnings, and price momentums, suggesting that revenue surprises, earnings surprises, and prior returns each carry some exclusive unpriced information content. Furthermore, the profits

of momentum driven by firm fundamental performance information (revenue or earnings) depend upon the accompanying firm market performance information (price) and vice versa.

Chen et al. (2016) examine how fundamental accounting information can be used to supplement technical information to separate momentum winners from losers. The authors introduce a ratio of liquidity of buy volume to liquidity sell volume (BOS ratio) to proxy the level of information asymmetry for stocks and show that the BOS momentum strategy can enhance the profits of the momentum strategy. They also propose a unified framework by incorporating the FSCORE (Piotroski 2000) and the GSCORE (Mohanram 2005) into the momentum strategy. The empirical results show that the combined investment strategy includes stocks with a larger information content that the market cannot reflect in time, and therefore, the combined investment strategy outperforms the momentum strategy by generating significantly higher returns.

Recently, Fu et al. (2024) examine the interplay between free cash flows (FCFs) and price momentum—both cross-sectional and time-series—in predicting future stock returns. They find that FCFs and past returns offer complementary and independent predictive power, suggesting that technical and fundamental signals jointly shape return dynamics. A strategy that conditions momentum on FCFs—longing past winners with high FCFs and shorting past losers with low FCFs—substantially outperforms traditional momentum strategies in terms of return, Sharpe ratio, and downside risk. The enhanced strategy's profitability is robust to investor sentiment, time variation, transaction costs, and is not explained by accruals, limits to arbitrage, or standard risk factors. Further decomposition shows that the return predictability is mainly driven by net cash distributions to equity and debt holders, highlighting investors' tendency to underreact to persistent, valuation-relevant components of cash flows due to an earnings fixation.

Overall, the evidence suggests that composite momentum strategies can generate significant positive returns and improve portfolio performance. However, developing and implementing these strategies can be challenging and requires careful consideration of the underlying variables and their interactions.

5 Source of momentum effects

The phenomenon of the momentum effect is intriguing and widely discussed in the literature because it challenges the weak-form Efficient Market Hypothesis (EMH), originally proposed by Fama (1970). According to the EMH, past information should have no predictive power regarding future price movements. However, the strong persistence of such return patterns over subsequent periods raises a fundamental question that has naturally attracted the interest of numerous scholars. Indeed, the rationales potentially explaining momentum are principally motivated in one of three ways: efficient (risk-based) explanations, pure underreaction to information, and continuing overreaction to information. In this section, we review related studies within each of these three frameworks.

5.1 5.1 Efficient explanations

Various researchers have proposed efficient explanations for the momentum effect. These explanations are rooted in the EMH, which suggests that stock prices should fully reflect all

available information (Fama 1970). This implies that strategies relying on past price trends, such as momentum, should not consistently yield abnormal returns. While Fama (1970) provides a theoretical framework against momentum effects, Fama (1991) suggests that certain anomalies in stock returns do not necessarily contradict market efficiency.

Carhart (1997) suggests that the momentum effect may result from market frictions, such as transaction costs, which prevent investors from quickly and efficiently incorporating new information into asset prices. Consequently, Carhart (1997) proposes a four-factor model by adding a momentum factor (MOM) to the Fama-French three-factor model:

$$r_{i,t} = \beta_i^0 + \beta_i^1\ MKT_t + \beta_i^2 SMB_t + \beta_i^3 HML_t + \beta_i^4 MOM_t + e_{i,t} \tag{4}$$

which includes market risk (*MKT*), size (*SMB*), book-to-market ratio (*HML*), and momentum (*MOM*) to explain stock returns. Studies have shown that the momentum factor can capture a distinct source of risk and return beyond market, size, and book-to-market factors.

Pástor and Stambaugh (2003) investigate whether market-wide liquidity is a state variable important for asset pricing. They extend the Fama-French three-factor model by adding a liquidity risk factor (L),

$$r_{i,t} = \beta_i^0 + \beta_i^1 MKT_t + \beta_i^2 SMB_t + \beta_i^3 HML_t + \beta_i^4 L_t + e_{i,t}. \tag{5}$$

Their study finds that expected stock returns are cross-sectionally related to the sensitivity of returns to fluctuations in aggregate liquidity. They find that a liquidity risk factor accounts for half of the profits of a momentum strategy over a 34-year period.

Liu and Zhang (2008) suggest that momentum could result from macroeconomic risks. Incorporating the factors of Chen et al. (1986)—including unexpected inflation (UI), changes in expected inflation (DEI), term premium (UTS), and the growth rate of industrial production (MP)—they estimate factor loadings to explore whether these influence momentum profits. Empirical evidence shows that more than half of momentum profits can be explained by macroeconomic risk factors, particularly the loading on MP.

Novy-Marx (2012) provides evidence that the traditional view of momentum—the tendency of a stock to persist in its current motion—does not accurately describe the return spreads between winner and loser groups. Instead, the evidence shows that intermediate-horizon past returns have greater predictive power than recent past returns. In Fama-MacBeth regressions, the coefficient on intermediate past returns is significantly higher than that on recent past returns. Moreover, momentum strategies based on intermediate past returns tend to earn larger and more significant profits.

Kelly et al. (2021) argue that time-varying risk compensation could capture short-term momentum and long-term reversal. They first estimate latent factors with time-varying factor loadings that depend on observable firm characteristics based on instrumented principal components analysis. They show that time-varying factor loadings on corresponding variables can capture reliable conditional risk premia, which cannot be explained by other factor models used in the literature. Their estimates of conditional risk exposure can explain a sizable fraction of momentum and long-term reversal returns.

Ehsani and Linnainmaa (2022) examine whether price momentum in individual stocks relates to momentum in factor returns. Following Kozak et al. (2020), they extract 47 principal component-based factors and find that factor momentum is concentrated in high-eigenvalue components—those with the strongest explanatory power for the cross-section of stock returns. A strategy trading the top 10 eigenvalue components generates a five-factor alpha with a t-statistic of 6.51. Moreover, momentum in these dominant factors transmits to stock-level momentum through time-varying factor loadings, offering a parsimonious explanation for the persistence documented by Jegadeesh and Titman (1993).

Cakici et al. (2025) offer a comprehensive global examination of the relationship between factor momentum and price momentum across 51 international equity markets. Their findings challenge prior claims, such as those by Ehsani and Linnainmaa (2022), that factor momentum subsumes price momentum. Specifically, while principal component (PC) factor momentum captures a substantial portion of price momentum profits in developed markets such as the U.S. and U.K., empirical factor momentum fails to consistently explain momentum returns across countries. In many cases, stock price momentum better explains factor momentum than vice versa, indicating that momentum remains a distinct and robust anomaly, not merely a byproduct of timing factor exposures. Their evidence also refutes claims that factor momentum can subsume industry momentum on a global scale, reinforcing the persistent and multifaceted nature of momentum effects.

Complementing this perspective, Ma et al. (2024) construct both time-series and cross-sectional factor momentum strategies in the Chinese stock market and find economically and statistically significant returns—annualized 9.91% with a Sharpe ratio of 1.15. Notably, they show that factor momentum subsumes stock momentum, high-priced stock momentum, and industry momentum in China, suggesting that factor-based return persistence is the dominant source of momentum anomalies in markets where stock-level momentum is weak or absent. Their analysis also reveals that factor momentum is stronger under high idiosyncratic volatility, elevated information asymmetry, and greater short-sale constraints, highlighting the role of mispricing correction and limits to arbitrage. Taken together, these findings reinforce the view that momentum operates through the dynamics of latent risk factors and is shaped by both the structure of the market and investor behavior.

While studies contend that the momentum effect aligns with the risk factors and does not contradict the efficient market hypothesis, Taylor (1982) counters the random walk hypothesis, pointing out the presence of price trends. This lends indirect support to momentum-based strategies, thereby complicating the weak-form market efficiency. Moreover, Brown (2010) introduces a critical viewpoint, suggesting that market anomalies, such as momentum, might not align with the conventional efficient market hypothesis, further enriching the debate.

In subsequent sections, we explore behavioral explanations, with several studies linking momentum to trading volume. Lee and Swaminathan (2000) propose a volume-based momentum life cycle hypothesis, while Hofstede (2001) associates individualism (proxied by trading volume) with momentum profits. Byun et al. (2016) find higher profits in stocks with continuing overreaction using a volume-informed measure. However, trading volume is not exclusively tied to behavior; it is also a vital factor in asset pricing. Lee et al. (2013) derive a simultaneous equation model showing a disequilibrium price adjustment process. Their findings, consistent with Campbell et al. (1993) and Lo and Wang (2000), suggest that

trading volume is an important factor in capital asset pricing, implying that its explanatory power for momentum could still stem from market efficiency.

5.2 Behavioral explanations

From a behavioral perspective, the momentum effect may be triggered by investor misreaction to information due to behavioral biases. Such an explanation can further be classified into two main categories: underreaction and overreaction phenomenon. The former originally refers to Barberis et al. (1998) and Hong and Stein (1999), while the latter refers to Daniel, Hirshleifer, and Subrahmanyam (1998). In contrast to traditionalists, behaviorists argue that traders may not always act rationally, and the market is also difficult to keep in efficiency because of arbitrage risk (Shleifer and Vishny 1997). Underreaction and overreaction effects are thus identified as the major sources of mispricing and the core concepts in behavioral finance. In this section, we briefly review related literature.

5.2.1 Underreaction

In Barberis et al. (1998), conservatism bias refers to the tendency of individuals to attach insufficient weight to new public information about a firm. This creates momentum through an initial market underreaction. However, the representativeness bias refers to individuals who receive a sequence of good (or bad) news become optimistic (or pessimistic) about firms' future value. As a result, firms experiencing prolonged periods of good news would become overvalued, and those experiencing long periods of bad news tend to become undervalued. The return patterns of these stocks ultimately reverse as realized information fails to meet expectations. Lee and Swaminathan (2000) support these views by proposing a volume-based Momentum Life Cycle hypothesis. They find that trading volume aims to reconcile short-term underreaction and long-term overreaction.

Hong and Stein (1999) alternatively develop a theoretical model populated by two groups of boundedly rational agents: "news watchers" and "momentum traders." The news watchers trade only on private information about fundamentals, whereas the momentum traders trade only on past price movements. News watchers who fail to extract other news watchers' information from prices make information diffuse gradually across the population, causing prices to underreact in the short run. Momentum traders thus can profit by trend-chasing through such an underreaction. As more and more momentum traders arrive in the market, the initial underreaction inevitably turns into an overreaction at longer horizons. In sum, Hong and Stein (1999) provide a context for reconciling the dynamics of short-term underreaction and long-term overreaction.

Investors' underreaction to information can also stem from various behavioral biases and market inefficiencies. Da et al. (2014) argue that momentum arises because of the inability or reluctance of investors to process all available information. Limited attention to information arriving in small but continuous information is akin to the "frog-in-the-pan" effect that underreacts as the water is slowly brought to a boil. To deal with this argument, they propose a proxy of information discreteness as below:

$$\mathrm{ID} = \mathrm{sgn}\left(\mathrm{PRET}\right) \times \left[\%\mathrm{neg} - \%\mathrm{pos}\right], \tag{6}$$

where PRET is defined as a firm's cumulative return over the past year after skipping the most recent month. The sign of PRET is denoted sgn(PRET) and equals + 1 if PRET > 0 and − 1 if PRET < 0. A large ID measure signifies discrete information, while a small ID measure signifies continuous information. The authors show that stocks with gradual accumulation of information exhibit more momentum than stocks with returns that are accumulated in a lumpy fashion.

Based on the "frog-in-the-pan" hypothesis, Büsing et al. (2024) decompose momentum returns into pre-peak and post-peak components. They find that over 80% of momentum profits originate from the pre-peak phase—when prices rise gradually before reaching their interim high—reflecting a persistent underreaction mechanism consistent with Da et al. (2014). This "forgotten" component not only avoids the crash risk typically associated with conventional momentum but also delivers higher Sharpe ratios, positive skewness, and robustness across market conditions and global equity markets, thereby offering a behaviorally grounded and empirically robust refinement of momentum-based strategies.

Building on these theoretical and empirical foundations, a substantial body of research has investigated the behavioral and structural sources of underreaction that contribute to the momentum effect. These mechanisms can be broadly categorized into: (1) informational frictions, including information asymmetry, information uncertainty; and (2) investor heterogeneity, including limited attention, heterogeneous beliefs, and gradual information diffusion; (3) limits to arbitrage; (4) cognitive biases, such as disposition effect, and anchoring; and (5) investor sentiment. The following discussion examines these factors in greater detail, highlighting how each contributes to the persistence of return momentum documented in the literature.

Blume, Easley, and O'Hara (1994) suggest that information asymmetry can create underreaction to new information in the market. Information asymmetry occurs when one party in a transaction has superior information compared to the other party, leading to underreaction by uninformed market participants. Jiang et al. (2005) and Zhang (2006) propose that information uncertainty can arise when market participants lack confidence in the accuracy or reliability of new information. If the slow market response to information is due to psychological biases such as overconfidence, these psychological biases will be larger and, hence, will be slower when there is more ambiguity about the implications of the information for a firm's value. The momentum effect thus becomes stronger as the degree of information uncertainty becomes greater.

Hong et al. (2000) test the gradual-information-diffusion model of Hong and Stein (1999). They show that gradual information diffusion refers to the slow dissemination of new information in the market, especially bad information; this leads to underreaction by investors. Momentum strategies thus work better among stocks with low analyst coverage or small size. These results are consistent with the hypothesis that firm-specific information diffuses only gradually across the investing public. Similar evidence can refer to Goyal et al. (2025) who provide a more comprehensive international evidence in support of Hong et al. (2000). Using firm-level data from both developed and emerging markets, they find that momentum is strongest when information arrives gradually rather than in large chunks—an implication of limited attention and slow information diffusion. Their findings also align with the "frog-in-the-pan" hypothesis of Da et al. (2014), showing that investors tend to underreact to small, continuous news updates. Notably, the proxy for information discrete-

ness is consistently significant across countries, confirming that underreaction to diffuse information is a robust source of momentum outside the U.S. context.

Hong and Stein (2007) further address three underlying mechanisms that generate heterogeneity among investors, namely gradual information flow, limited attention, and heterogeneous priors. They suggest that investors may overly simplify stock price behaviors as simple functions of some macro variables when they do not notice the other important variables. Once the investors use differential priors (i.e., heterogeneous priors or beliefs) or assessments of the data (i.e., the "likelihood") to assess a firm, the heterogeneity among them persists and leads to large price reactions. Forecast errors estimated in any overly simplified model will be persistent; this leads to momentum. Both Scheinkman and Xiong (2003) and Verardo (2009) suggest that heterogeneous beliefs can lead to underreaction and contribute to the momentum effect.

Limits to arbitrage—such as short-selling constraints, transaction costs, and limited capital—impede arbitrageurs' ability to exploit mispricing, thereby allowing anomalies like momentum to persist. Ali and Trombley (2006), Arena et al. (2008), and D'Avolio (2002) argue that such frictions contribute to delayed price adjustments and underreaction. Extending this view, Atilgan et al. (2020) demonstrate that momentum profits are amplified among stocks characterized by high left-tail risk, limited investor attention, and elevated arbitrage costs—particularly when retail investors overprice these stocks. Recent evidence further underscores the importance of investor composition in explaining return continuation. Du et al. (2025) further find that momentum is substantially stronger in high-nominal-price stocks in China, where round-lot trading rules limit retail participation. This setting reveals that momentum is more detectable when noise trading—often associated with unsophisticated investors—is attenuated. Their cross-market validation across countries with similar institutional constraints reinforces the role of investor heterogeneity and trading frictions in shaping momentum profitability.

Abhyankar et al. (2024), however, investigate the role of option markets in mitigating short-sale constraints that hinder momentum arbitrage. They show that the decline in momentum profits over the past two decades is largely attributable to the expansion of stock options trading, which serves as a substitute for costly or restricted short selling. In particular, the erosion of momentum profitability is concentrated in the short leg of the strategy, where options facilitate more efficient price correction. Their findings underscore how arbitrage limits—especially those related to shorting—are central to the persistence of momentum returns.

Another behavioral explanation for momentum can be attributed to Grinblatt and Han (2005), who argue that the disposition effect causes this pattern. The disposition effect refers to the tendency of investors to sell winning positions too early and hold losing positions too long, leading to underreaction to positive news. In their model, a reluctance to sell losers and an eagerness to sell winners both cause prices to underreact to true fundamental news for losing stocks. They show that momentum is related to unrealized capital gains in their setting.

George and Hwang (2004) demonstrate that if traders consider available price information, the 52-week high price, they would underreact to new information when the current stock price close to the 52-week high price. People use the 52-week high as an "anchor" when assessing the increment in stock value implied by new information. Nearness to the

52-week high dominates the forecasting power of past returns for future returns and explains a large portion of the profits from momentum investing. Chen et al. (2024) further find that when a stock's price is close to its 52-week high, investors tend to anchor on this reference point, resulting in insufficient reaction to recent news. This anchoring-based underreaction is especially pronounced in high-turnover stocks, where differing investor opinions further delay full price adjustment. Their findings reinforce the view that short-term momentum arises not solely from news arrival but also from behavioral frictions in interpreting price signals.

Finally, investor sentiment has been documented as one of the sources that could explain the momentum effect. In particular, Stambaugh et al. (2012) consider a setting with the presence of market-wide sentiment and short-sale impediments and find that several price anomalies are associated with high sentiment periods, including momentum strategy. This result suggests a richer understanding of how sentiment plays a role in pricing financial assets. Antoniou et al. (2013) alternatively consider another explanation for the conflict in reaction to the news, called cognitive dissonance. Cognitive dissonance refers to the discomfort felt by individuals when faced with conflicting beliefs or information, leading to underreaction to new information in the market. That is, investors react properly to the news which confirms their beliefs but underreact to reports that disconfirm their beliefs. On average, therefore, they underreact, which gives rise to momentum. They show that momentum arises principally in optimistic periods because investors underreact to bad news.

5.2.2 Overreaction

In contrast to Barberis et al. (1998) and Hong and Stein (1999), Daniel, Hirshleifer, and Subrahmanyam (1998) argue that traders initially overreact to new information; as a result, momentum occurs. Daniel, Hirshleifer, and Subrahmanyam (1998) attempt to explain patterns using investors' overconfidence and self-attribution. They suggest that overconfident investors assess arriving public signals with a self-serving bias that overweights private signals and underweights public signals, indicating a continuing overreaction followed by a slow correction. Long-term reversals are the results of long-run corrections to prior to overreaction.

Daniel, Hirshleifer, and Subrahmanyam (1998) and Hong and Stein (1999) also suggest that investors may initially underreact to information because they fail to process it fully or are reluctant to update their beliefs based on limited signals. Gutierrez and Prinsky (2007) extend this idea by examining how delays in overreaction depend on the type of information investors receive. Daniel and Hirshleifer (2015), Jiang et al. (2005), and Scheinkman and Xiong (2003) further argue that investor overconfidence leads to overreaction to news, generating momentum profits. Zhang (2006) shows that investor overconfidence is higher during periods of elevated market volatility, during which momentum profits are also stronger.

Continuing overreaction, on the other hand, refers to the persistence of past price trends into the future. Byun et al. (2016) show that momentum profits are higher for stocks with continuing overreaction. They argue that this is because investors overreact to the initial news and continue to do so for an extended period due to overconfidence, leading to strong price trends that persist. They propose a volume-based continuing overreaction measure to capture the trend in overconfidence,

$$CO_{i,t} = \frac{sum(w_J \times SV_{i,t-J}, \ldots, w_1 \times SV_{i,t-1})}{mean(Vol_{i,t-J}, \ldots, Vol_{i,t-})}$$
$$SV_{it} = \begin{cases} Vol_{i,t}, & if \quad r_{i,t} > 0 \\ 0, & if \quad r_{i,t} = 0 \\ -Vol_{i,t}, & if \quad r_{i,t} < 0 \end{cases}, \tag{7}$$

where $Vol_{i,t}$ is the dollar trading volume of stock i in month t, which is calculated as the sum of daily trading volume within the month, and $r_{i,t}$ is the stock i's return in month t. The results are stronger among stocks primarily held by investors more prone to biased self-attribution. Their results provide direct support for the model of return predictability based on overconfidence and biased self-attribution.

Several studies further show that investor overconfidence intensifies during favorable market conditions. Cooper et al. (2004) argue that momentum profits arise from systematic differences in returns across market states and are higher when markets remain in an upward state due to investor optimism. Asem and Tian (2010) further show that market-state transitions play an important role in driving momentum profits, with overconfidence strengthening when markets persist in the same direction.

Chui et al. (2010) examine how cultural differences influence the returns of momentum strategies. By using the individualism index developed by Hofstede (2001), they find that individualism is positively associated with trading volume and volatility as well as the magnitude of momentum profits. The main reason behind this result is that, in high-individualism countries, investors are overconfident and believe that they can identify and exploit profitable trading opportunities. They, therefore, demonstrate that momentum profits are higher for stocks that are more actively traded, indicating greater individualism among investors.

A recent paper by Hou et al. (2025) further refines the overreaction explanation by linking it to selective investor attention. They develop a behavioral model in which limited attention leads to asymmetric information processing. When attention is directed toward past price movements rather than fundamentals, investors overreact to price signals, generating price momentum. Their empirical results show that price momentum is primarily driven by overreaction mechanisms and is distinct from the underreaction underlying earnings momentum.

Overall, behavioral explanations suggest that momentum profits arise due to the tendency of investors to underreact or overreact to information. The specific sources of underreaction or overreaction may vary, depending on the type of information, the market state, or individual investor characteristics.

5.2.3 Momentum life cycle

Lee and Swaminathan (2000) propose the momentum life cycle hypothesis to explain momentum profits. Using turnover as a proxy for trading volume, they document an asymmetric volume effect in stock returns. Price momentum is more pronounced among high-volume stocks, in which loser stocks primarily drive this relation. Specifically, the momentum life cycle framework posits that low-volume (high-volume) firms exhibit many characteristics associated with value (glamour) stocks. As a result, early-stage momentum strategies—constructed by taking long positions in low-volume winners and short positions in high-volume losers—are more likely to persist, whereas late-stage momentum strategies

composed of high-volume winners and low-volume losers are more likely to reverse quickly. In this framework, the early stage corresponds to underreaction followed by an overreaction phase, while the late stage relates to price correction. The novelty of the momentum life cycle hypothesis lies in its ability to explain the coexistence of short-term underreaction and long-term overreaction by explicitly incorporating trading volume and past returns.

The momentum life cycle theory has received empirical support. Jegadeesh and Titman (2001) find that momentum profits are concentrated in the first few months following an initial price trend, suggesting that early-stage momentum profits are driven by underreaction. Similarly, Hong and Stein (2007) show that momentum profits are related to the number of investors following a stock, consistent with the positive-feedback trading mechanism emphasized by the momentum life cycle hypothesis.

Lee and Swaminathan (2000) argue that existing behavioral theories do not fully account for all of their evidence. Models proposed by Daniel, Hirshleifer, and Subrahmanyam (1998), Barberis et al. (1998), and Hong and Stein (1999) capture the spirit of the momentum life cycle hypothesis by allowing prices to initially underreact and eventually overreact to fundamental news. However, none of these models explicitly incorporate trading volume and therefore cannot fully explain why volume predicts the magnitude and persistence of future price momentum. Lee and Swaminathan (2000) suggest that investors' limited attention may help explain this relation. Nevertheless, the finding that momentum intensifies as the trading volume of loser stocks increases remains difficult to reconcile fully within the momentum life cycle framework. D'Avolio (2002) argues that higher trading volume reflects greater investor disagreement, which in turn contributes to mispricing.

Overall, the momentum life cycle hypothesis provides a complementary explanation to both efficient-market and behavioral accounts of momentum. By emphasizing how market reactions evolve over time in response to price movements and trading volume, the momentum life cycle framework helps explain why momentum profits tend to emerge in the short run but attenuate or reverse over longer horizons.

6 The business cycle, regime transitions, and momentum performance

A substantial body of literature documents that momentum profitability varies systematically across economic conditions and market regimes. Early evidence by Chordia and Shivakumar (2002) shows that momentum profits are concentrated in expansionary periods and largely disappear during economic downturns, suggesting that macroeconomic conditions play an important role in shaping momentum performance. Building on this insight, recent studies further refine the link between momentum strategies and the business cycle by explicitly modeling regime dependence.

A prominent contribution in this direction is Goulding et al. (2023), who decompose time-series momentum signals into slow-moving (12-month) and fast-moving (1-month) components to identify four distinct market states—Bull, Bear, Correction, and Rebound. They show that Bear states, characterized by negative slow and fast signals, are concentrated in the early stages of recessions and are associated with negative expected returns and elevated downside risk. Importantly, their analysis provides a new interpretation of momentum crashes, linking them to adverse regime transitions rather than to unconditional tail risk. By blending fast and slow signals, intermediate-speed momentum strategies are

shown to adapt more effectively to trend reversals and deliver superior Sharpe ratios. Their decomposition of performance further demonstrates that abnormal returns arise not only from static factor exposures, but also from dynamic adaptation to changing market regimes.

Beyond business-cycle dynamics, investor sentiment has long been identified as a key force amplifying momentum effects. Hong et al. (2000) propose a model in which information diffuses gradually across investors with heterogeneous sophistication, generating momentum through delayed reactions. In this framework, elevated investor sentiment increases belief dispersion and strengthens trend-chasing behavior. Empirical evidence supports this mechanism. Stambaugh et al. (2012) show that momentum profits are significantly stronger during periods of high investor sentiment and weaken substantially when sentiment is low. Similarly, Baker et al. (2012) document higher momentum profits during optimistic sentiment regimes across international markets. Related evidence from options markets, such as Li et al. (2018), further suggests that volatility expectations and speculative demand interact with sentiment to shape momentum profitability, particularly during turbulent periods.

7 Momentum crashes

While momentum profits tend to be amplified during favorable economic and sentiment conditions, they are also exposed to severe downside risk. A central insight of the momentum literature is that momentum crashes are not random tail events, but are highly predictable and closely tied to volatility and panic states. Barroso and Santa-Clara (2015) show that the risk of momentum strategies varies substantially over time and can be accurately forecast using realized volatility. By scaling momentum exposure inversely with its conditional volatility, they propose a volatility-managed momentum strategy that virtually eliminates momentum crashes while nearly doubling the Sharpe ratio relative to the conventional approach. Their findings imply that momentum crashes reflect time-varying risk rather than inherent instability of momentum strategies.

This view is further developed by Daniel and Moskowitz (2016), who demonstrate that momentum crashes tend to occur in panic states—typically following market declines, during periods of elevated volatility, and contemporaneously with sharp market rebounds. Crucially, they show that both the conditional mean and conditional variance of momentum returns are predictable. A dynamically weighted momentum strategy based on these forecasts substantially improves performance, approximately doubling both alpha and the Sharpe ratio compared to a static momentum strategy. Their analysis firmly establishes momentum crashes as a regime-dependent phenomenon rather than an unavoidable feature of momentum investing.

Complementing these findings, Moreira and Muir (2017) generalize volatility-based scaling across a wide range of asset pricing factors. By adjusting factor exposures according to aggregate market volatility, they show that risk-managed strategies achieve higher Sharpe ratios and lower drawdowns, highlighting the broader relevance of volatility timing in mitigating crash risk. Related evidence from industry momentum strategies shows that the benefits of volatility-based risk management remain robust even in settings where momentum exhibits limited time-varying beta, underscoring that effective crash mitigation does not hinge on a specific market structure (Grobys et al. 2018).

Hanauer and Windmüller (2023) further advances the momentum literature by comparing three enhanced momentum strategies designed to mitigate momentum crashes and improve risk-adjusted returns across both U.S. and international markets. These strategies include constant volatility-scaled momentum, constant semi-volatility-scaled momentum, and dynamic-scaled momentum. Utilizing a large dataset spanning 49 equity markets from 1926 to 2017, the authors find that all three approaches substantially increase Sharpe ratios while reducing skewness, kurtosis, and maximum drawdowns relative to standard momentum strategies. Notably, the dynamic-scaled momentum strategy exhibits the lowest sensitivity to time-varying investor overconfidence, suggesting a superior risk management profile. The study further documents that momentum returns are closely linked to market continuation states, consistent with prior findings that momentum profits tend to be higher during persistent bull or bear markets. These enhanced strategies demonstrate greater robustness and implementability than traditional momentum, with stronger performance concentrated outside of January, underscoring the importance of volatility scaling in improving momentum profitability and stability.

Recent work further refines this perspective by explicitly isolating crash states from normal momentum regimes. Dierkes and Krupski (2022) show that momentum crashes systematically occur in rebounding bear markets characterized by high momentum volatility and negative momentum betas. Using ex-ante crash indicators based on both systematic and momentum-specific risk measures, they demonstrate that crash periods can be effectively separated from momentum bull markets. A trading strategy that conditions momentum exposure on these indicators substantially reduces drawdowns and delivers higher risk-adjusted performance than existing volatility-managed approaches, reinforcing the view that momentum crashes are predictable and state-dependent events rather than rare tail realizations.

To sum up, the literature suggests that momentum strategies exhibit strong regime dependence. Momentum profits tend to be higher during expansions and optimistic sentiment regimes, but are also vulnerable to sharp losses during periods of market stress and transition. Importantly, a growing body of evidence shows that momentum crashes are largely predictable and can be effectively mitigated through volatility-based risk management, dynamic scaling, and explicit crash-state identification. These findings underscore that the economic relevance of momentum strategies depends not only on their average returns, but also on their exposure to time-varying crash risk and the effectiveness of risk management mechanisms.

8 Short-horizon return dynamics: momentum, reversals, and investor clienteles

Stock returns often exhibit short-term reversals that gradually transition into momentum over longer horizons. While much of the momentum literature attributes this horizon dependence to behavioral biases or delayed information diffusion, a growing strand of research emphasizes the role of investor clienteles, trading horizons, and information flow in shaping short-horizon return dynamics. From this perspective, whether returns display short-term reversal or short-term momentum depends not only on the nature of information, but also on who trades, how frequently information arrives, and which type of investor dominates trading activity.

Recent empirical evidence strongly supports this view. Medhat and Schmeling (2022) document the existence of "short-term momentum" at monthly horizons in U.S. equities, particularly among large, liquid, and high-turnover stocks. They show that these stocks—typically dominated by active institutional investors—exhibit return continuation rather than reversal at short horizons, consistent with more frequent information releases and faster trading responses. In contrast, stocks with lower turnover and greater retail participation tend to display short-term reversals, suggesting that price pressure from uninformed or liquidity-motivated trading plays a more prominent role in those segments.

Related evidence is provided by Chui et al. (2022), who exploit institutional differences between Chinese A- and B-share markets. They find that B shares—characterized by greater institutional ownership and more informed trading—exhibit momentum but not short-term reversals, whereas A shares—dominated by retail investors—display strong short-term reversals and weak momentum. This sharp contrast within the same firms and economic environment highlights the central role of investor composition in determining short-horizon return patterns.

A theoretical foundation for these findings is developed by Jegadeesh et al. (2025). Using a multiperiod model with short- and long-horizon noise traders alongside active investors who underreact to information they do not originate, they show that short-term reversals and longer-term momentum can arise endogenously from the interaction of heterogeneous traders. In their framework, reversals reflect temporary price pressure induced by short-horizon noise traders, while momentum emerges from gradual trading responses to new information. Importantly, the model predicts that the prevalence of reversal versus momentum, and the horizons over which these patterns appear, depend on investor composition and the frequency of information arrivals rather than on systematic mispricing.

This investor-clientele perspective complements earlier evidence linking short-term reversals to retail trading and liquidity demand. For example, Kaniel et al. (2008) show that individual investors tend to buy after price declines and sell after price increases, generating short-term reversals, while Nagel (2012) links short-horizon return reversals to liquidity-driven trading and volume shocks. Together, these studies reinforce the view that short-term return dynamics are shaped by the interaction between investor types and trading horizons.

Overall, this strand of literature suggests that short-term momentum and short-term reversal should not be viewed as competing anomalies, but rather as outcomes that emerge under different investor compositions and information environments. When active, informed investors dominate and information arrives frequently, short-horizon momentum is more likely to arise; when trading is driven by short-horizon noise traders or liquidity demand, short-term reversals tend to prevail. This investor-clientele channel provides a microstructure-based explanation for horizon-dependent momentum that is distinct from behavioral bias explanations based on underreaction or overreaction, and offers a useful lens for interpreting heterogeneous momentum patterns across assets and markets.

9 Momentum effect in cryptocurrency market

The rapid expansion of cryptocurrency markets has prompted a growing body of research examining whether momentum—one of the most robust return regularities in financial markets—extends to digital assets. Early evidence delivers mixed conclusions. Some studies

document economically meaningful momentum profits, while others report pronounced short-term reversals, leading to an apparently fragmented literature.

Several early papers provide direct evidence of short-horizon momentum in cryptocurrencies. Using weekly data, Grobys and Sapkota (2019) show that winner–loser portfolios earn positive abnormal returns, although the effect dissipates quickly as holding horizons lengthen. Similarly, Tzouvanas et al. (2020) document significant momentum profits at short horizons and show that such strategies can offer diversification benefits when combined with traditional assets. These findings suggest that return continuation is present in cryptocurrency markets, but is concentrated at short horizons. In contrast, other studies emphasize reversal effects, particularly at very short horizons. Using broad samples of cryptocurrencies, Li et al. (2019) and Shen et al. (2020) find strong short-term reversals, especially when daily data are employed. These reversal patterns appear difficult to reconcile with the momentum evidence at first glance and have fueled debate over whether cryptocurrencies exhibit systematic return continuation at all.

An important insight offered by Zaremba et al. (2021) is that much of this apparent disagreement arises from differences in empirical design rather than from fundamentally contradictory evidence. Reviewing a wide range of studies, they show that results depend critically on sample composition and horizon choice. Studies focusing on a narrow set of economically significant cryptocurrencies—those dominating market capitalization and trading activity—are more likely to document momentum, whereas studies employing expansive universes that include many small and illiquid tokens tend to find reversals. This distinction is economically meaningful, as smaller cryptocurrencies are characterized by thin liquidity, extreme volatility, and noisy price dynamics. Differences in formation and holding horizons further contribute to heterogeneity. While some studies attempt to replicate the classic equity momentum framework of Jegadeesh and Titman (1993) using multi-month ranking periods, others focus on weekly or daily horizons that may be more appropriate for a young asset class. As summarized by Zaremba et al. (2021), daily return dependence in cryptocurrencies often manifests as reversal, whereas momentum—when present—tends to emerge at short weekly horizons. Once these design choices are taken into account, much of the conflicting evidence can be reconciled.

Building on this foundation, Liu and Tsyvinski (2021) provide a broad characterization of cryptocurrency risk–return patterns and report a strong time-series momentum component in major cryptocurrencies, highlighting that trend-following predictability is a salient feature of this asset class. They also emphasize that cryptocurrency returns are forecastable by market-specific forces, including proxies related to user adoption and attention, reinforcing the view that momentum-like return continuation in cryptocurrencies may be closely tied to belief formation and investor behavior rather than slow-moving fundamentals.

A more unified cross-sectional perspective is developed by Liu et al. (2022), who embed cryptocurrency momentum in a factor-pricing framework. Using weekly data, they document robust short-horizon cross-sectional momentum and construct a cryptocurrency momentum factor that earns a significant premium alongside market and size factors. Importantly, their results indicate that short-term momentum is not merely a statistical artifact of particular samples, but a systematic component of expected cryptocurrency returns that helps organize a wide set of return predictors within a coherent factor structure.

Recent evidence also suggests that “momentum” in cryptocurrencies can be understood more generally as exposure to trends at different horizons and with different signal construc-

tions. Fieberg et al. (2025) propose a trend factor for the cross section of cryptocurrency returns, showing that trend-based signals capture priced variation in expected returns even when momentum is implemented through technical, moving-average-style measures. This line of work strengthens the broader conclusion that short-horizon continuation in cryptocurrencies is not confined to a single portfolio construction, but appears across related implementations of trend and momentum.

More recent research further shows that cryptocurrency momentum is highly conditional. Rather than depending solely on recent returns, momentum profitability depends on how market conditions evolve over time. Using a transition-based framework following Asem and Tian (2010), Hsieh et al. (2025) show that momentum profits arise only in persistent up-market regimes and disappear during down markets or state reversals. This evidence complements earlier findings by demonstrating that even within economically significant cryptocurrencies, momentum is path-dependent and sensitive to regime continuity.

Taken together, the momentum effect in cryptocurrency markets is short-lived, heterogeneous, and strongly context-dependent. Early disagreements largely reflect differences in asset coverage and horizon choice, while more recent studies reveal that momentum is further shaped by market regimes and investor behavior. Cryptocurrencies therefore provide a valuable setting for reassessing the scope and limits of momentum theories, particularly those emphasizing behavioral mechanisms and belief dynamics.

10 Conclusion

Momentum-based investment strategies represent one of the most robust and extensively documented anomalies in empirical finance. Originating from behavioral theories—particularly those emphasizing investor underreaction, limited attention, and extrapolative expectations—momentum challenges the core tenets of market efficiency and continues to provoke theoretical debate and practical innovation. A wealth of empirical evidence demonstrates that momentum strategies consistently generate abnormal returns across a broad array of asset classes, including equities, government and corporate bonds, foreign exchange, and commodity markets. These return patterns persist across geographies and time periods, suggesting that the momentum premium may be driven by deep-seated behavioral biases or slow-moving institutional frictions that are not easily arbitraged away.

Yet, momentum is not a free lunch. Despite its apparent profitability, momentum investing is subject to well-documented risks—particularly its susceptibility to abrupt reversals during market regime shifts, heightened volatility, or flight-to-safety episodes. These "momentum crashes" can result in significant drawdowns, undermining the strategy's risk-adjusted performance and requiring sophisticated portfolio construction techniques (e.g., dynamic risk management or volatility scaling) to mitigate tail risk.

Notwithstanding these challenges, momentum strategies remain a cornerstone of modern quantitative investing. Their relatively low correlation with traditional value-based or macroeconomic factor strategies enhances portfolio diversification, while their implementation has been facilitated by advances in data processing, machine learning, and real-time analytics. Today, momentum signals are widely employed by hedge funds, asset managers, and rules-based ETFs, underscoring their enduring practical relevance.

However, it is essential to recognize that momentum strategies are not universally appropriate. Their effectiveness is highly contingent on investor-specific constraints—such as liquidity needs, turnover tolerance, tax considerations, and behavioral disposition toward losses. Hence, the application of momentum should be contextualized within a broader portfolio framework and aligned with clearly defined investment objectives and risk parameters.

In conclusion, momentum investing offers both theoretical intrigue and empirical credibility. When properly understood and cautiously implemented, momentum can serve as a powerful tool for generating excess returns, enhancing risk-adjusted performance, and capturing inefficiencies left behind by behavioral biases and market imperfections. Future research should continue to refine our understanding of the mechanisms that sustain momentum, its interaction with other asset pricing factors, and its evolving role in increasingly complex and adaptive financial markets.

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Declarations

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